Filtered Virtual Reference: A New Method for the Reduction of Power Line Interference With Minimal Distortion of Monopolar Surface EMG

Alberto Botter* and Taian M. Vieira

***Abstract*—*Goal:* This study tests and validates a new method to remove power line interference from monopolar EMGs detected by multichannel systems: the filtered virtual reference (FVR). FVR is an adaptation of the virtual reference (VR) method, which consists in referencing signals detected by each electrode in a grid to their spatial average. Signals may however be distorted with the VR approach, in particular when the skin region where the detection system is positioned does not cover the entire muscle. *Methods:* Simulated and experimental EMGs were used to compare the performance of FVR and VR in terms of interference reduction and distortion of monopolar signals referred to a remote reference. *Results*: Simulated data revealed the monopolar EMG signals processed with FVR were significantly less distorted than those filtered by VR. These results were similarly observed for experimental signals. Moreover, FVR method outperformed VR in removing power line interference when it was distributed unevenly across the signals of the grid. *Conclusion:* Key results demonstrated that FVR improves the VR method as it reduces interference while preserving the information content of monopolar signals. *Significance:* Although the actual distribution of motor unit action potential is represented in monopolar EMGs, collecting high quality monopolar signals is challenging. This study presents a possible solution to this issue; FVR provides undistorted monopolar signals with negligible interference and is insensitive to muscle architecture. It is therefore relevant for EMG applications benefiting from a clean monopolar detection (e.g., decomposition, control of prosthetic devices, motor unit number estimation).**

***Index Terms*—Average reference, common mode, electromyography, power line interference, signals, virtual reference.**

I. Introduction

HIGH-DENSITY surface electromyography based on 2-D electrode arrays is becoming common in research laboratories and attracting clinical interest [1]–[5]. A problem inherent to this as well as to other techniques (EEG, ECG) is the removal of power line interference caused by parasitic capacitive coupling with the power line. Such interference is expected to generate a similar contribution to all monopolar channels of the detection system, constant in space and variable in time at the power line frequency (50 or 60 Hz) and its harmonics. It would therefore appear that a method to reduce power line interference could be based on the differential measurement of the voltage between each electrode of the array and the instantaneous spatial average of all the detected signals. This method is referred to as "virtual reference" (VR) and has been widely used in ECG and EEG applications [6].

The VR method is based on two assumptions. The first assumption is that, for any given time instant, the spatial integral on a surface that covers the entire spatial support of the potential generated by the active sources is zero [7]. Under this assumption, the fluctuations of the spatially averaged signal represent only the external common mode voltage that can be removed by subtraction. However, this approximation is valid only if the detection system covers the entire spatial support of the active sources [8], [9]. For EEG applications this requirement is approximately fulfilled by high-density detection systems, with electrodes covering the whole scalp. The application of this technique to high-density EMG recordings can be more problematic because the surface integral of the potential can be considered null only when the detection system covers the entire muscle surface and goes beyond the tendon regions. Since this condition can rarely be achieved during practical EMG recordings, the application of the VR technique might imply a distortion of both propagating and nonpropagating components of monopolar EMG signals. This effect is described, for the specific case of signals detected with a linear array of electrodes, in [2].

The second assumption on which the VR method is based is that the external interference affects all the channels of the detection system with the same amplitude and phase. If this hypothesis is not valid, the average signal might not be representative of the interference affecting individual channels, and the VR method may not remove the power line interference from all channels in the detection system [2].

Based on the VR technique, in this study, we proposed a new method to remove power line interference with marginal distortion of the actual, biological signal; filtered virtual reference (FVR) method. First, a theoretical account of the VR spatial filtering effect on the bidimensional potential distribution is presented. Then, simulated and experimental EMGs are used to test for whether the FVR technique: 1) minimizes the signal distortion associated with the VR processing (i.e., due to the removal of the average of the monopolar signals); and 2) accounts for the different amplitudes of the interference affecting the individual channels of the detection system.

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II. Methods

A. Analysis of VR Technique

In EMG recordings, monopolar signals detected over a skin parallel-fibered muscle include both propagating and nonpropagating components. Nonpropagating components are mainly due to crosstalk and to the potential generation (at the neuromuscular junctions) and extinction (at the muscle-tendon junctions, often referred as end-of-fiber effect) and may contribute strongly to the EMG signal [10].

If the spatial support of these potentials is not entirely covered by the detection system, the nonpropagating components do not have instantaneous zero spatial average. Consequently, the subtraction of the spatial average affects both the nonpropagating and propagating components of the monopolar signals [2]. Moreover, when the potential distribution is sampled with a grid of electrodes, the nonuniform spatial distribution of the surface potential (e.g., the transversal decay in case of skin parallel-fibered muscles) is expected to be an additional factor degrading the quality of the signals detected with the VR technique. In this section, we generalize the analysis presented in [2] to the case of bidimensional sampling of the potential distribution.

Let us consider a simple model with a grid of electrodes placed between the endplate locations (innervation zone) and the tendons with *M* columns, parallel to the fibers' direction, and *N* rows [see Fig. 1(a)]. The origin of the reference system corresponds to the electrode $(r, c) = (1, 1)$. $V(x, z, t)$ is a potential distribution, on the skin surface above the muscle, travelling with the same amplitude in the *z*-direction with a conduction velocity *v*. Under these conditions, the potential distribution generated by a fiber can be written as $V(x, z, t) = a(x) \cdot V(z, t)$ where *a(x)* is the function ($\in$[0, 1]) that describes the decay of the potential in the transverse direction (*x*).

For a generic column *c* (*x* coordinate $= x_c$), the potential distribution in one spatial dimension (*z*) is $a_c \cdot V(z, t)$ with $a_c = a(x_c)$. For a given time instant (t_o) the voltage detected by the electrode of row *r* (coordinate z_r) and column *c* (coordinate x_c) using the VR method is

$$V_{\mathrm{VR}}\left(x_c, z_r, t_0\right) = a_c \cdot V\left(z, t_0\right) \cdot \delta\left(z - d \cdot r\right) - V\left(z, t_0\right) \cdot \frac{1}{N \cdot M} \sum_{k=1}^{M} \sum_{i=1}^{N} a_k \cdot \delta\left(z - d \cdot i\right) \quad (1)$$

where *d* is the distance between two consecutive electrodes along the fiber direction (interelectrode distance), and *δ(.)* is the Dirac distribution. The second term of (1) is the spatial average computed over the *N* rows ($i = 1, \ldots, N$) and the *M* columns ($k = 1, \ldots, M$) of the detection system.

The equivalent spatial impulse response (2) and the transfer function in time frequency domain (3) are

$$h_{\mathrm{VR}}^{r,c}(z) = \frac{V_{\mathrm{VR}}\left(x_c, z_r, t_0\right)}{a_c \cdot V\left(z, t_0\right)} = \delta\left(z - d \cdot r\right) - \frac{1}{a_c \cdot N \cdot M} \cdot \sum_{k=1}^{M} a_k \cdot \sum_{i=1}^{N} \cdot \delta\left(z - d \cdot i\right) \quad (2)$$

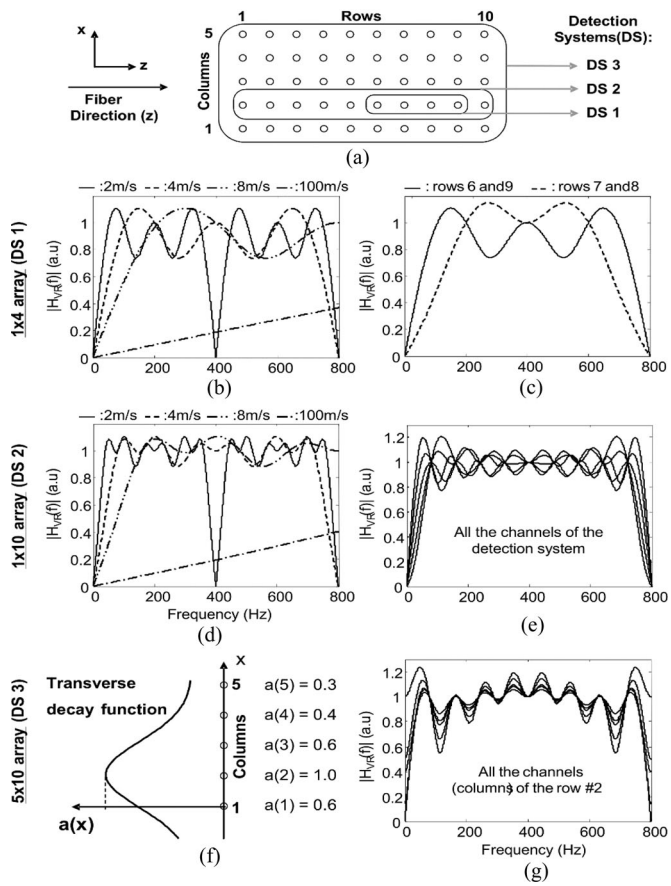

Fig. 1. Theoretical analysis of the VR method (Section A, Methods). (a) Schematic representation of three detection systems: two linear arrays with four and ten electrodes (DS 1 and DS 2, respectively) and a grid of 5 × 10 electrodes (DS3) placed on one side of the innervation zone (monodirectional propagation). (b) DS 1: magnitude of the transfer function $H_{\mathrm{VR}}(f)$ of the channel in row 9, column 2 (channel 4 of DS1) for different conduction velocities (*v*) of the travelling potential $V(z, t)$. (c) Magnitude of $H_{\mathrm{VR}}(f)$ for the four channels of DS1 considering a conduction velocity (*v*) of 4 m/s. (d) DS 2: magnitude of $H_{\mathrm{VR}}(f)$ for the channel in row 9, column 2 (channel 9 of DS 2) for different conduction velocities (*v*) of the travelling potential $V(z, t)$. (e) Magnitude of $H_{\mathrm{VR}}(f)$ for the ten channels of DS 2 ($v = 4$ m/s). (g) DS 3 : magnitude of $H_{\mathrm{VR}}(f)$ for all the channels (columns) of the row 2 considering a signal propagating with a conduction velocity (*v*) of 4 m/s. H_{VR} *(f)* of panel (g) was computed considering the transversal decay function - *a(x)* - depicted in panel (f).

$$H_{\mathrm{VR}}^{r,c}(\omega) = e^{-j\frac{\omega}{v}dr} - \frac{\sum_{k=1}^{M} a_k}{a_c \cdot N \cdot M} \cdot \frac{1 - e^{-j\frac{\omega}{v}dN}}{1 - e^{-j\frac{\omega}{v}d}} \quad (3)$$

where ω/v is the spatial angular frequency. The transfer function depends on the three following parameters: 1) the conduction velocity (*v*) of the propagating signal [see Fig. 1(b) and (d)]. Indeed, it can be noted that the common mode components of the signal ($v \to \infty$ and $a_k = 1$ for any *k*) are removed by the VR method ($|H_{\mathrm{VR}}| = 0$, for $v \to \infty$); 2) the frequency (*f*) of *V(z, t)* and the distance (*d*) between two consecutive electrodes; 3) the number of rows (*N*) of the detection system. In addition to the aforementioned considerations described in [2], the analysis presented in this section indicates that when the signals are detected with bidimensional grids of electrodes, the transfer function of VR is influenced also by the transversal (along the

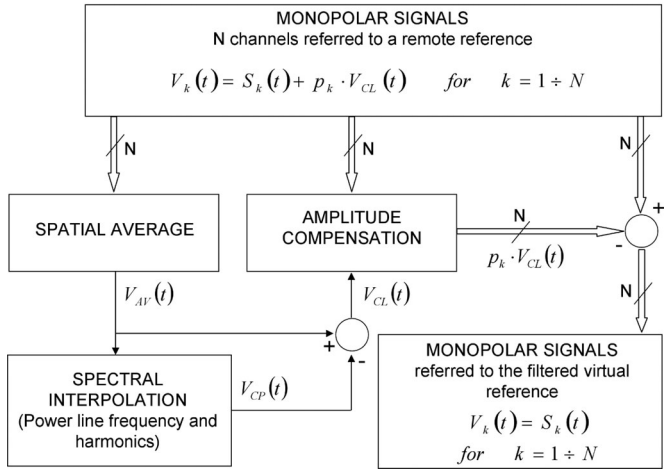

Fig. 2. Block diagram of the FVR method. V_k is the monopolar signal recorded by the electrode *k* of a detection system with *N* electrodes; S_k is the interference-free monopolar signal; $V_{CL}(t)$ is the power line interference; p_k is the scaling factor taking into account possible variations of the interference amplitude across the channels; $V_{CP}(t)$ is the common mode signal exclusively due to physiological components; $V_{AV}(t)$ is the spatial average calculated over the detection system, including the contribution of power line interference ($V_{CL}(t)$) and of physiological components ($V_{CP}(t)$).

columns) amplitude profile [see Fig. 1(f)]. Considering the second term of (3) depends on the transverse decay function (*a*), $|H_{\mathrm{VR}}|$ will be different for the different columns of the detection system for signal components whose amplitude is not constant in space.

According to these theoretical considerations, the VR technique may greatly distort the actual EMGs and may not remove the power line interference from all channels of the grid. The magnitude of such distortion cannot be anticipated, as it depends on different factors (e.g., conduction velocity, dimension of the detection system, etc.; see Fig. 1). To circumvent these issues, we therefore propose some adaptations to the VR method.

B. FVR Method

FVR is an adaptation of the VR method and was designed to remove the power line interference from surface EMGs with minimal distortion of the actual, monopolar signals. The main features of this method are: 1) it considers only the power line component of the common mode signal (i.e., the spatial average), and 2) it accounts for differences in the amplitude of the power line interference affecting the different channels of the detection system.

The block diagram of the FVR method is reported in Fig. 2. Let us consider the monopolar signal (V_k) recorded by an electrode *k* of a detection system constituted by *N* electrodes (4). V_k can be written as the sum of the actual monopolar signal referred to a remote reference ($S_k(t)$) and the power line interference ($V_{CL}(t)$). Variations in the amplitude of the power line interference across channels are represented in (4) by multiplying $V_{CL}(t)$ by a scaling factor ($p_k \in [0, 1]$).

$$V_k(t) = S_k(t) + p_k \cdot V_{CL}(t). \tag{4}$$

According to the VR method, the second term of (4) is estimated through the spatial average of all the signals of the detection system ($V_{AV}(t)$).

As shown in the previous section, if the detection system does not cover the entire spatial support of the motor unit action potentials (MUAPs), both the power line interference ($V_{CL}(t)$) and the physiologically-induced ($V_{CP}(t)$) common mode (e.g., standing waves) will contribute to $V_{AV}(t)$. FVR therefore aims at separating $V_{CL}(t)$ from $V_{CP}(t)$; this operation is performed through the blocks "*Spatial Average*" and "*Spectral Interpolation*" reported in Fig. 2. "*Spatial Average*" computes $V_{AV}(t)$ and the block "*Spectral Interpolation*" linearly interpolates the power line frequency and its harmonics in $V_{AV}(t)$ by considering a 2-Hz interval centered on the interpolated spectral frequencies [11]. This procedure removes the power line components ($V_{CL}(t)$) from $V_{AV}(t)$ and provides an estimation of the physiological contribution to the common mode ($V_{CP}(t)$). $V_{CL}(t)$ is then obtained by subtracting $V_{CP}(t)$ from $V_{AV}(t)$. The second step of the method (block "*Amplitude Compensation*") corresponds to the estimation of the scaling factor p_k for each channel of the detection system. Under the assumption that $S_k(t)$ and $V_{CL}(t)$ are not correlated ($<S_k(t), V_{CL}(t)> = 0$)

$$<S_k(t) + p_k \cdot V_{CL}(t), V_{CL}(t)> = p_k \cdot <V_{CL}(t), V_{CL}(t)>. \tag{5}$$

Thus, the scaling factor p_k can be estimated as

$$p_k = \frac{<V_k(t), V_{CL}(t)>}{<V_{CL}(t), V_{CL}(t)>}. \tag{6}$$

The block "*Amplitude Compensation*" computes p_k for each channel of the detection system thus providing a set of *N* reference signals ($p_k \cdot V_{CL}(t)$ for $k = [1 \text{ to } N]$). According to (4), the interference-free monopolar signals ($S_k(t)$) are obtained, for each channel *k*, by subtracting the new reference signal from each detected signal, $V_k(t)$.

C. Simulation Study

In this section, high-density EMGs were simulated to investigate the effect of the detection system size and of the localized distribution of muscle activity on monopolar EMGs referred to a remote reference, to the VR and to the FVR. This simulation study was focused on the quantification of signal distortion associated with the rereference procedure (either VR or FRV) rather than on the efficacy in removing power line interference, which is addressed in the following section.

The simulations were based on a model of recruitment of a population of MUs [12] and on a cylindrical volume conductor model [13]. The model was implemented in Matlab software (R2007a, The MathWorks Inc., MA, USA). The cylindrical volume conductor consisted of an anisotropic muscle layer, an isotropic subcutaneous tissue, a skin layer and a muscle with elliptical, physiological cross-sectional area (transversal x depth, 30 × 25.4 mm). The innervation zones of the motor units were located at the center of the muscle (see Fig. 3). The parameters of the anatomical model are reported in Table I.

Ten contractions with constant excitation levels were simulated, ranging from 10% to 100% of the excitation level in steps

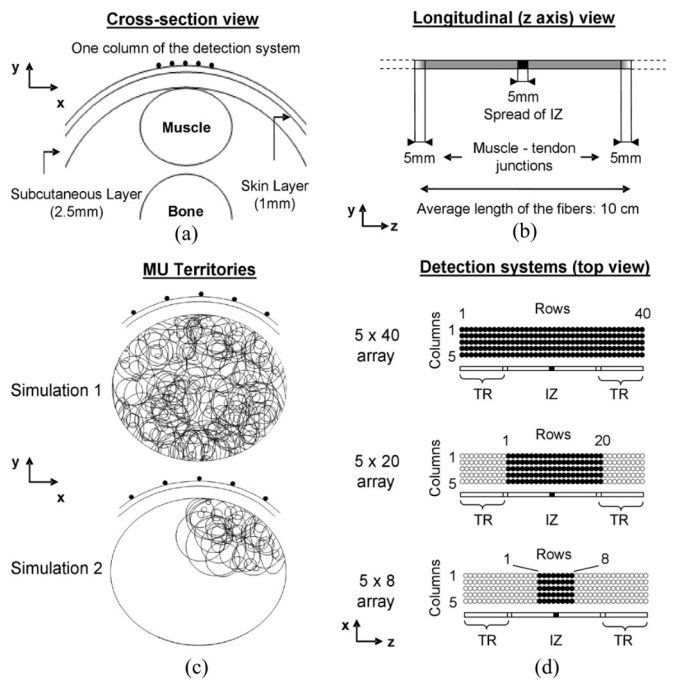

Fig. 3. (a) Cross-sectional view of the anatomical model. It consisted of bone, muscle tissue, a subcutaneous layer, and skin. The black circles over the skin layer represent one row (five electrodes) of the detection system. (b) Longitudinal view (z-axis) of one representative fiber. (c) Territories of the simulated motor units. In "Simulation 1" the territories of the active motor units were distributed randomly throughout the muscle, whereas in "Simulation 2" they were located close to the skin surface. (d) Schematic representation of the three detection systems used, each arranged around the center of the simulated fibers (innervation zone location). For each detection system a schematic representation of the simulated fibers is shown (IZ: innervation zone, TR: tendon region).

TABLE I
PARAMETERS OF THE CYLINDRICAL MODEL

	Model Parameter	Value
Motor unit properties	Fiber density*	~20 fib/mm^2
	Innervation ratio**	25–1500
	Discharge rate	8–30 pps
	CoV of interspike interval	15%
Muscle properties	Number of motor units	200
	Number of muscle fibers	15 000
	Number of MU fibers (range)	15–1500
	Muscle PCSA	598 mm^2
	Average fiber length	100 mm
	Tendon ending spread	5 mm
	Innervation zone spread	5 mm
Limb properties	Skin thickness	1 mm
	Subcutaneous tissue thickness	2.5 mm
	Bone (radius)	20 mm

of 10%. The simulated monopolar signals were detected at 4096 samples/s with circular electrodes (diameter 1 mm), arranged in a grid with five columns and 40 rows (200 electrodes) with 5-mm interelectrode distance in both the longitudinal and transverse directions (see Fig. 3). The center of the grid corresponded to the center of the muscle volume projected on the skin surface. The detection system covered both the muscle (approximately 20 electrodes corresponding to the central portion of each column) and the tendon regions (approximately ten electrodes over each tendon).

The electrodes of the 5×40 array were grouped in subsets to obtain different detection systems [5×20 array and 5×8 array, see Fig. 3(d)]. Each detection system was arranged around the center of the 5×40 array.

The surface potential was simulated as detected by the three detection systems in two conditions. In the first condition (Simulation 1) the territory of active motor units was distributed randomly throughout the muscle [see Fig. 3(c)], whereas in the second condition (Simulation 2) the active motor units were a subpopulation of superficial motor units [see Fig. 3(c)]. Simulation 1 and 2 were considered to compare VR and FVR when the active motor units generate either a uniform (Simulation 1) or localized (Simulation 2) distribution of muscle activity under the electrodes. Indeed, as demonstrated in the first section (see Fig. 1), a steep transversal decay of the potential (present in Simulation 2) is a possible source of error in VR signals.

For each of the two conditions, ten randomly selected motor-unit populations were simulated. These simulations were sought to represent ten muscles, with the same anatomical characteristics but with different random locations of the motor units within their volume.

For each detection system and simulation condition, VR and FVR method were applied to the simulated signals. The comparison between the original monopolar signal (referred to a remote reference) and the signals referred to the virtual (VR) and filtered virtual (FVR) references was performed for each channel with the normalized mean square error (NMSE), according to the formula

$$NMSE_{i,j} = 100 \cdot \frac{\sum_{n=1}^{N} \left[f_{i,j}(n) - g_{i,j}(n)\right]^2}{\sum_{n=1}^{N} f_{i,j}(n)^2} \tag{7}$$

where $f_{i,j}(n)$ is the monopolar signal referred to a remote reference; $g_{i,j}(n)$ denotes the signal referred to the virtual (or filtered virtual) reference; i and j are the indexes of the channel (row and column respectively), and N is the number of samples of the signal.

D. Experimental Signals

The performance of the VR and FVR methods was assessed on signals acquired with different detection systems and from muscles with different architectures (i.e., biceps brachii and forearm muscles) to provide examples of the effect of VR and FVR in real, experimental conditions.

High-density surface EMG signals were acquired in monopolar derivation (referred to a remote reference), amplified, bandpass filtered (3-dB bandwidth, 10–500 Hz), sampled at 2048 Hz and A/D converted with a resolution of 12 bits (multichannel surface EMG amplifier, EMG-USB, OT Bioelettronica, Torino, Italy).

Two experiments were performed in one subject to address specific issues. In the first experiment (EXP 1) we focused on the effect of VR and FVR on nonpropagating components detected through a linear array positioned along the muscle fiber direction in a skin parallel-fibered muscle. Since the nonpropagating components in these signals do not have instantaneous zero

spatial average, they are expected to be removed by VR, but preserved by FVR. In the second experiment (EXP 2) we tested FVR and VR on monopolar signals with a marked transverse decay; as predicted in section *A* and *B*, VR but not FVR is expected to distort the monopolar signals and therefore their spatial amplitude distributions. The possibility of accounting for uneven power line interference across channels with the FVR approach is tested for the signals acquired in both experiments. The protocols considered in the two experiments are briefly described below.

EXP 1: EMG signals were detected from the short biceps brachii (BB) head with a linear electrode array of 16 electrodes and 10-mm interelectrode distance (SPES Medica, Battipaglia, Italy) during an isometric contraction at 30% MVC. The electrodes were aligned parallel to the muscle fibers, approximately covering the proximal and distal BB tendons.

EXP 2: EMG signals were detected from dorsal, forearm muscles with 96 electrodes arranged in a grid of 12 rows and eight columns with 10-mm interelectrode distance (LISiN, Politecnico di Torino, Torino, Italy). Signals were recorded during isometric wrist extension at 10% of the maximum voluntary contraction (see [14] for details). Images were created from the root mean square (RMS) values computed for raw monopolar EMGs and for EMGs conditioned with both VR and FVR methods. Changes in the spatial distribution of EMG amplitude were then assessed.

For both the experiments, a set of bandpass filters centered at power line frequency and its first five harmonics was used to extract the power line interference from monopolar EMGs recorded in rest conditions. Power line interference was multiplied by a scaling factor and added to the EMG channels to test how robustly the VR and the FVR perform in the presence of nonuniformly distributed power line interference. In EXP 1 the scaling factor was linearly increased (in the range 0–5) from channel 1 to channel 16. In EXP 2, the scaling factor was 5 for the six proximal channels and 0 for the six distal channels. All EMG signals were analyzed with Matlab software (R2007a, The MathWorks Inc., MA, USA).

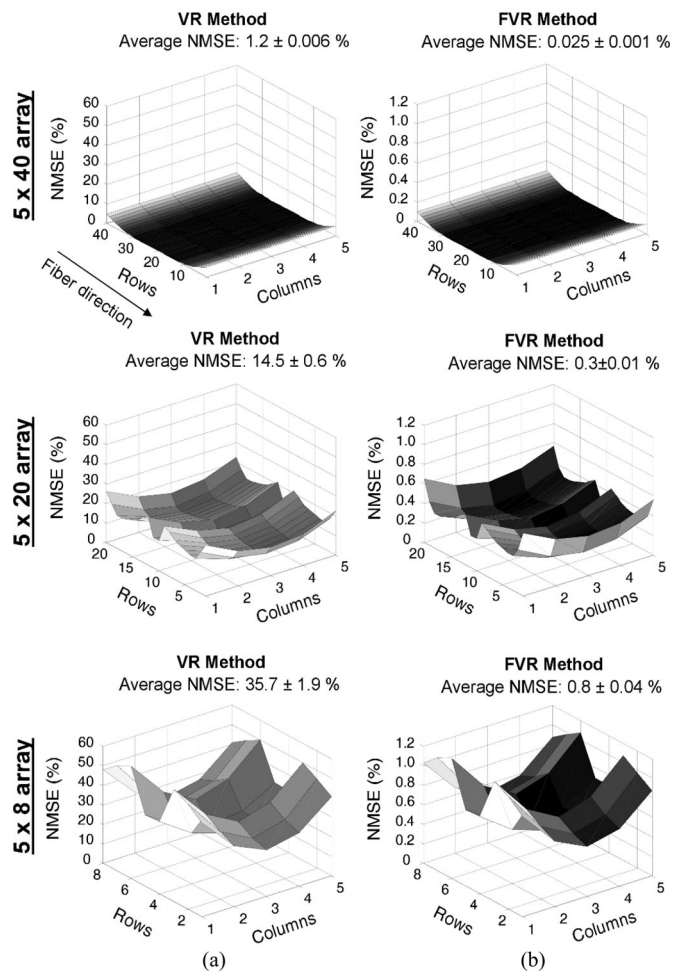

Fig. 4. Comparison between FVR and VR method for one representative set of signals (Simulation 1, 10% MVC) recorded by three detection systems. For each detection system, the original set of signals (monopolar referred to a remote reference) was referred to the VR and to the FVR. The NMSE between the original signals and the VR or FVR set of signals was computed to compare both methods (panels (a) and (b) respectively). The 3-D representation of NMSE and the average NMSE computed over the electrodes of each detection system is reported. Note different z-scale between panels (a) and (b).

E. Statistical Analysis

A three-way ANOVA (10 contraction levels x 3 detection system sizes x 2 reference method) was applied to compare the NMSE values of simulated signals. Parametric statistics were considered after ensuring the data distribution obtained for the ten populations of motor units simulated was Gaussian (Lilliefors test). Bonferroni correction was considered for the pair-wise comparisons, performed whenever ANOVA indicated a significant, additive effect. Results are given as mean $\pm$ std.

III. Results

A. Simulation 1: Effect of VR and FVR Methods on Interferential Signals From Uniformly Distributed MUs

The contraction level simulated did not affect the performance of the two methods tested. Independently of whether simulating small or large electrode grids, for EMGs processed both with the VR and the FVR methods, we did not observe statistically significant differences among NMSE values obtained for the ten effort levels tested (from 10% to 100% MVC at 10% MVC steps ANOVA additive effect; $P > 0.3$ for all cases). For this reason, simulation results showing the effect of the size of the detection system on NMSE values are specifically reported for the 10% MVC condition.

For both VR and FVR methods, the size of the detection system affected the NMSE averaged over the grid as well as the distribution of NMSE values across electrodes. NMSE values distributed relatively less uniformly across channels in the grid as the size of the detection system simulated decreased (cf., rows in Fig. 4), with NMSE values being smaller for electrodes located along the central column and the central row. The NMSE averaged over all channels in the grid decreased significantly when either VR or FVR techniques were applied to EMGs simulated for larger detection systems (ANOVA additive effect;

$P < 0.001$). Regardless of the detection system size, as shown in Fig. 4, the mean NMSE value obtained for the FVR method was about 50 times smaller than that obtained for the VR method (note different z scales between panels (a) and (b) in Fig. 4; ANOVA additive effect; $P < 0.001$).

B. Simulation 2: Effect of VR and FVR Methods on Interferential Signals From Nonuniformly Distributed MUs

For MUs nonuniformly distributed within the muscle volume, the transverse decay of the potential critically affected the performance of the virtual and FVR methods. This effect is clearly shown for a single potential. Fig. 5 shows examples of simulated signals [Simulation 2, see Fig. 3(c)] detected with three detection systems (5×40, 5×20 and 5×8 electrode array). The NMSE values obtained for both VR and FVR methods increased for EMGs simulated with a progressively smaller detection system (cf., potentials from the left to right panels in Fig. 5). For EMGs detected closer to either border of the simulated grid (column 1 or 5), NMSE values were larger than those obtained for EMGs detected centrally in the grid (see Fig. 5), where the center of the MU territory was simulated. The performance of VR and FVR methods showed however a different degree of dependence on the size of the detection system. Regardless of whether simulating EMGs with small or large electrode grids, NMSE values obtained for the VR method was dramatically ($\sim$40 times) greater than those obtained for the FVR method (see Fig. 5). The considerations made for the single MUAP shown in Fig. 5 apply for the ten populations of 200 MUs simulated (ANOVA additive effect for the method and detection system size; $P < 0.001$).

C. Experimental Study: Distortion of the Signal

Crucial differences between VR and FVR were observed at the level of single motor units as well as for the spatial distribution of EMG amplitude. An example of a single MUAP for the BB muscle is shown in Fig. 6 (EXP 1 in methods). The original signal (monopolar referred to a remote reference) contains both propagating and nonpropagating components associated with the end-of-fiber effect. These components changed marginally when EMGs were processed with the FVR technique, whereas VR processing modified the shape of the propagating potential and markedly reduced the amplitude of end-of-fiber component (see Fig. 6). Similarly, when considering the distribution of the RMS amplitude of interference EMGs the VR and FVR methods provided different results (EXP 2 in methods). During isometric, wrist extension contraction, the RMS amplitude map obtained with the FVR method was similar to the amplitude map obtained for the raw, monopolar EMGs; in both cases greater EMGs were detected at the centro-lateral forearm region [see Fig. 7(a) and (b)]. The RMS amplitude of VR processed EMGs distributed in a substantially different fashion across the grid of electrodes, with largest EMGs being represented at the most distal row of electrodes [see Fig. 7(c)].

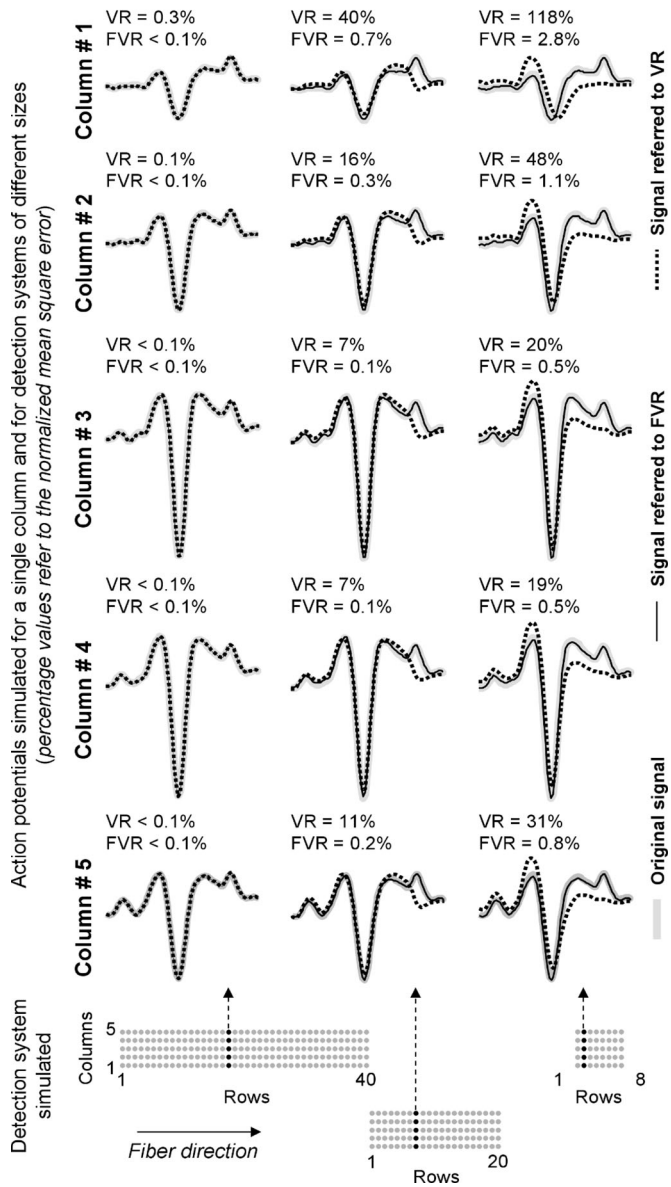

Fig. 5. Effect of the transversal decay of the potential. Example of simulated signals (Simulation 2) detected with three recording systems (5×40, 5×20 and 5×8 electrode array). Each row represents the signals detected by each channel of the selected row (see the schematic representation of the detection systems in the left part of the figure). For each channel the original signal referred to a remote reference (monopolar signal), the signal referred to the virtual reference (VR signal), and the signal referred to the FVR (FVR signals) are superimposed. The NMSE of VR and FVR signals are reported for each channel.

D. Experimental Study: Reduction of Power Line Interference

The two methods performed differently with respect to the attenuation of power line interference. Fig. 8(a) reports an example of BB monopolar surface EMG signals with power line interference added nonuniformly across electrodes (EXP 1 in methods). While the VR technique removed power line interference mainly from the central channels in the array (from channel 7 to channel 11; cf., Fig. 8(a) and (b), power line interference was equally suppressed across the whole array without

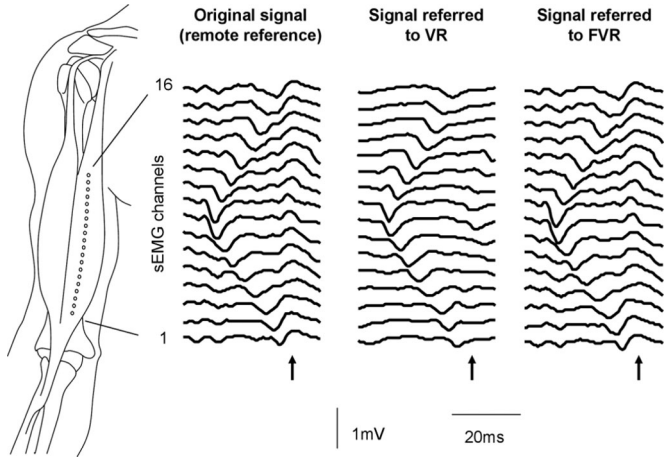

Fig. 6. Experimental surface recordings of an action potential measured from the biceps brachii during a 30%MVC isometric contraction (EXP 1). Original signals (monopolars referred to a remote reference), signals referred to the VR and to the FVR are shown. The modification of the propagating potential shape and the cancelation of the end-of-fiber component (indicated by the arrows) are visible for signals processed with VR. Both signal components are preserved in monopolar signals referred to FVR.

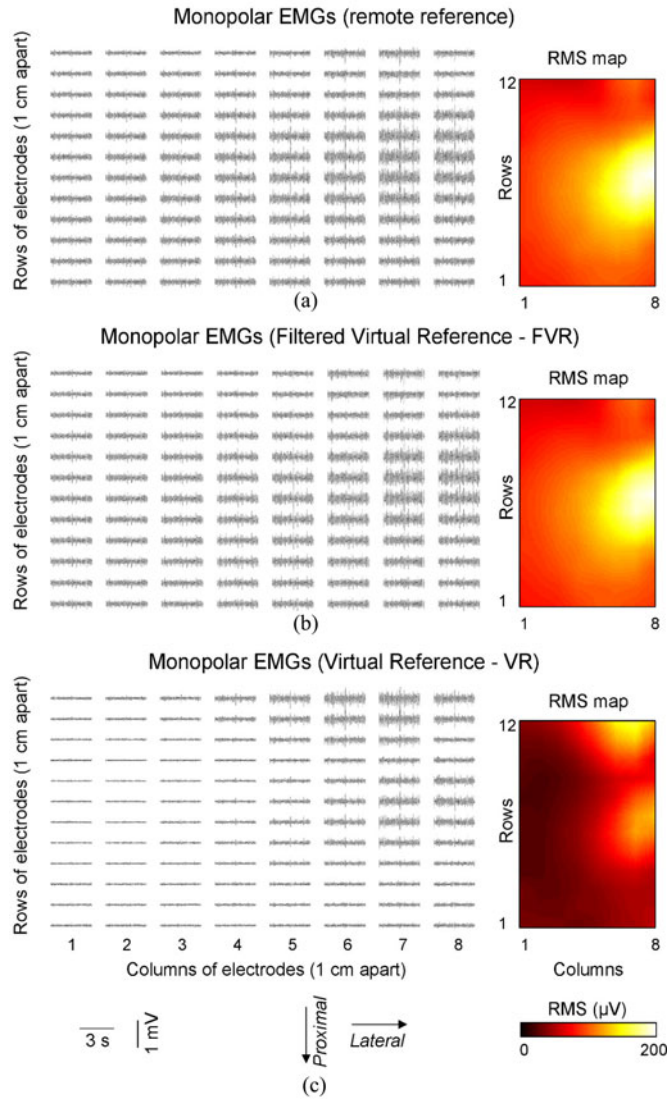

Fig. 7. EMG amplitude distributions over the dorsal portion of the forearm during a 20%MVC isometric wrist extension. RMS amplitude maps were computed over a 1-s epoch from monopolar signals referred to a remote reference (a) the FVR (b) and the VR (c) are reported.

changing the monopolar potentials with the FVR method [see Fig. 8(c)]. The same behavior was observed in the example reported in Fig. 8(d)–(f) (EXP 2 in methods) where the interference affected only a subset of the channels of the detection system. Therefore, regardless of whether muscle activity was detected along the whole detection system (EXP 1) or locally (EXP 2), and of how differently the power line interference affected the different channels in the detection system, the FVR method attenuated the power line interference in all channels without distorting the actual, monopolar signal.

IV. Discussion

The performance of the VR technique, commonly used to remove power line interference from multichannel recordings [6], and of the modified VR method proposed here, the FVR, is tested on simulated and experimental EMG signals. Both techniques are computationally advantageous with respect to other filtering techniques (e.g., empirical mode decomposition [15], [16]; spectral interpolation of individual channels [11]), as they apply to the signals' spatial average rather than to individual signals. This advantage may be particular attractive for EMG recordings based on a high number of electrodes. As discussed below, nevertheless, VR and FVR perform differently, both in terms of the amount of distortion of the actual, monopolar signals and of their efficacy in removing the power interference.

A. Filtering of Common Mode Signals With the VR: Distortion of Surface EMGs

The theoretical analysis of VR method showed that it behaves like a filter whose features depend on the conduction velocity of the propagating signals [see Fig. 1(b) and (d)] and on the position and size of the detection system [see Fig. 1(c) and (e)]. Whenever the common mode signal distributes unevenly across the grid of electrodes, the VR method distributes its energy across channels in the grid, greatly distorting the actual signals (see Figs. 4–8). Moreover, as discussed below, the VR method makes no distinction between the different sources contributing to the common mode signal.

Regardless of the size of the detection system, the VR method removes the common mode from EMG signals when it distributes equally across electrodes (see (3): $H_{\mathrm{VR}} = 0$, for $v \rightarrow \infty$). The components that contribute to the common mode are mainly two: the power line interference and the physiological nonpropagating potentials (standing waves) [2], [10]. Standing waves may be associated with the activity of a distant muscle (crosstalk [17], [18]), with the generation of MUAPs or with their extinction in the muscle-tendon junctions [10], [19]–[21]. One may argue that removing these standing waves from the EMG recordings is desirable, given that, for example, common

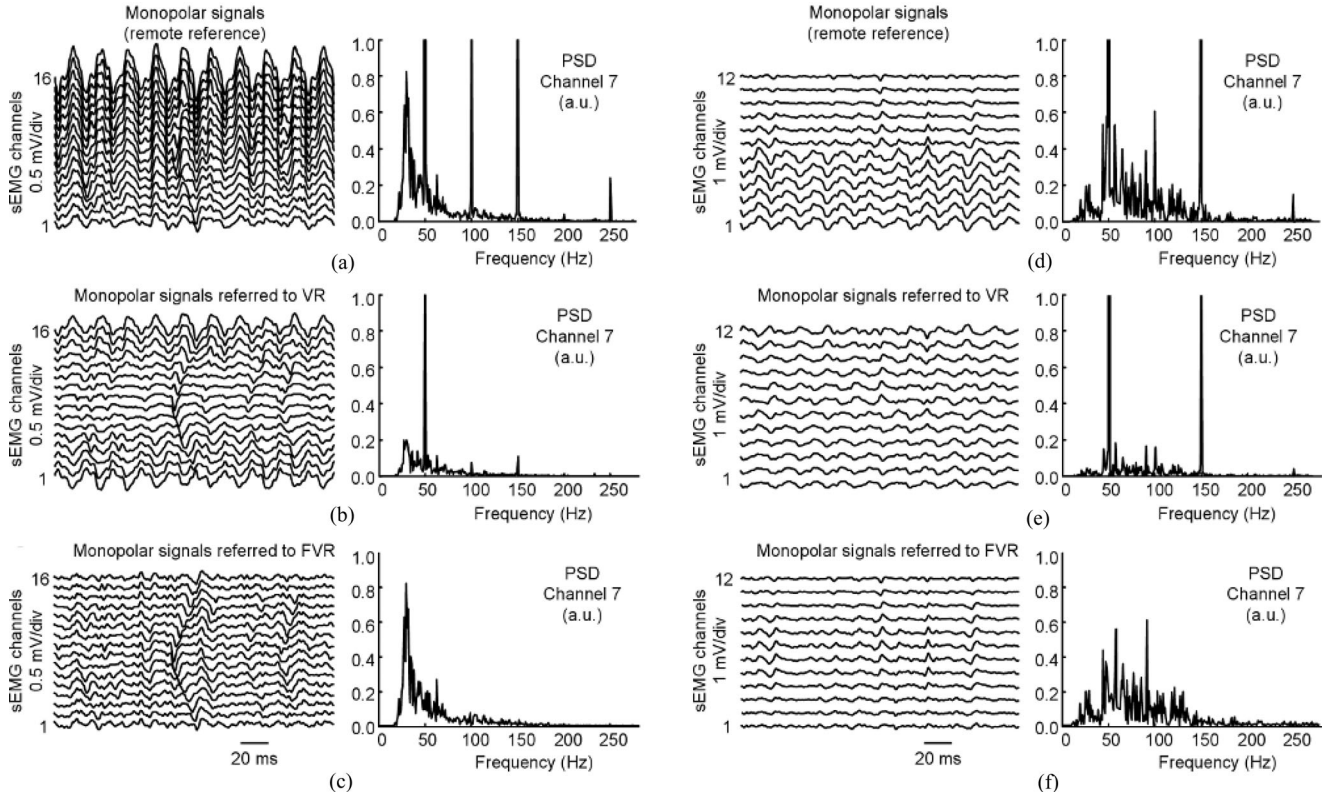

Fig. 8. Example of 16 monopolar signals detected longitudinally from the BB muscle (a)-(c) and from dorsal, forearm muscles (d)-(f). Signals shown in (a) and (d) were detected with respect to a remote reference, with artificial power line interference (50 Hz and four additional harmonics) added respectively to all signals with linearly increasing intensity increasing (from ch 1 to ch 16) and to a subset of signals (from ch 1 to ch 5). (b) and (e). Signals referred to VR. Note the reduction of physiological common mode signals. (c) and (f). Signals referred to FVR. The power spectral distribution of signals detected in ch 7 is shown alongside all panels to better illustrate the performance of the two methods. For each experiment, spectra are represented with the same scale (a.u.).

mode signals lead to overestimates of muscle fiber conduction velocity [22]. For such a purpose, however, one could rely on the use of selective spatial filters (e.g., single- and double-differential [19]) rather than on offline, filtering techniques. It must be noted that the method proposed here was strictly developed to attenuate the power line interference while preserving the standing waves in the monopolar EMGs.

When the amplitude of the common mode signal does not distribute uniformly across electrodes, the VR method greatly distorts the actual, monopolar signals (see Figs. 7 and 8). Occurrences of nonuniform spatial distribution of common mode signals may be observed both for power line interference and standing waves. In the case of power line interference, an uneven amplitude distribution could be associated with poor contact of some electrodes of the detection system [23] or with their geometrical position with respect to the interference source. Instead, the circumstances leading to nonuniform distribution of standing waves' amplitude are relatively more difficult to predict, as they depend on the muscle architecture and on the muscle size in relation to the size of the electrode grid. For example, considering that the amplitude of physiological, common mode sources decays with the inverse of their distance from electrodes [10], [24] (innervation zone and tendon regions), standing waves with different amplitude across electrodes are more likely to be observed for larger than smaller grids. Indeed, uneven distribution of standing waves' amplitude has been observed experimentally for muscles with long fibers (e.g., biceps brachii: [25]; forearm muscles: [14]), for muscles whose fibers attach to a broad bone region (e.g., trapezius: [26]) and for muscles with pennate geometry (e.g., vastus medialis: [27]; gastrocnemius: [28]). In either case, for power line interference and standing waves, the contribution of the channels detecting higher common mode potential will be distributed to all channels by the subtraction of the average: the VR approach. It is straightforward that this leads to a noncomplete cancelation of the power line interference and to a distortion of the physiological nonpropagating components of the EMG signal, with potentially crucial consequences for applications such as force estimation [29].

B. Robustness of the Proposed FVR Method: Preserving EMG Waveform

The FVR method was conceptually designed to remove the power line interference from surface EMGs without removing the standing waves and without distorting the actual, monopolar signals. Two main features potentially accounts for the robustness of the method proposed here: 1) it considers the power line component of the average signal and not other common mode sources, i.e., the standing waves; 2) it accounts for differences in the amplitude of the power line, common mode signal detected

by different channels in the grid. Results presented here, indeed, indicate that our proposed method successfully attenuates the power line interference, regardless of where it is detected in the grid. Most importantly, our findings show the FVR method does not distort the monopolar EMGs in the presence of standing waves represented with equal (see Fig. 6; EXP 1 in methods) and different (see Fig 7; EXP 2 in methods) amplitude in different channels; it is insensitive to spatial inhomogeneities associated with the physiological, common mode sources.

In addition to distortion due to common mode sources, distortion may arise for MUAPs when surface EMGs are detected by relatively small grids of electrodes. Small grids, with closely spaced electrodes, may not cover a skin region sufficiently large to include the whole spatial distribution of MUAPs. In this case, the spatial average of such potential distribution is not zero and thus the distortion resulting from the VR filtering does not uniquely apply to common mode components. Indeed, while the FVR technique retains the waveform of MUAPs, resulting in markedly small mean square error between the actual and the filtered signal (see Fig. 4), the VR method does not (cf., waveforms obtained using the VR and FVR methods for simulated and experimental signals in Figs. 5 and 6, respectively). The degree of distortion of MUAPs resulting from the VR filtering is variable. As shown in Fig. 1(b) and (d), the shape of VR transfer function changes with the conduction velocity of MUAPs. The dependence of the filter features on different conduction velocities and on the position of the detecting electrode decreases with the increase of the size of the detection system [see the reduction of $H_{\mathrm{VR}}(f)$ amplitude oscillations for comparisons between Fig. 1(b) and (d) and between Fig. 1(c) and (e)]. On one hand, increasing the size of the detection system may result in less distortion due to the VR filtering of MUAPs. On the other hand, as discussed above, the larger the detection system the more likely the standing waves are distributed unevenly across electrodes.

Another crucial point is the size and location of muscle activity represented in the surface EMGs. For large electrode arrays, covering muscles with pinnate geometry or a group of small muscles, EMGs with large amplitude may be detected from different, localized skin regions [14], [27], [30]. For example, largest EMGs were observed over different forearm regions when subjects were asked to isometrically extend their different finger extensor muscles [14]. Regardless of whether contracting any of the finger extensors, EMGs with largest amplitudes were detected by $\sim$20 out of 128 electrodes [14]. As predicted by our theoretical analysis and simulation results, and as suggested experimentally (see Fig. 7), such a localized representation of muscle activity may spread across all electrodes in the grid when EMGs are filtered with the VR rather than with the FVR method. Therefore, as the FVR method preserves the information content of the actual, monopolar EMGs, it does not demand any a priori knowledge on the spatial pattern of muscle activity.

An important remark of the FVR is that it exclusively applies to monopolar signals referred to a remote reference. Once EMGs are detected with the VR approach, the distortion it may introduce to the actual monopolar signals can be not accounted for by the FVR method. Therefore, whenever using a grid of electrodes with limited pick-up volume in relation to the muscle size or sampling from highly locally active muscles with a large grid, for example, detecting monopolar signals with the VR technique may be not appropriate.

C. Removing Power Line Interference Without Distorting EMGs

The VR method was not effective in reducing the power line interference from all the channels of the detection system as was the FVR method. This is due to the fact that the common mode (computed as the spatial average of all the signals of the detection system) is not representative of the interference affecting individual channels. More specifically, in both examples shown in Fig. 8 the amplitude of the power line interference is not the same for each channel in the grid and thus its subtraction removes the interference only in some locations and not in others. Moreover, since the detection systems did not cover the entire spatial support of the active sources, surface EMG signals were spatially filtered, as evidenced by the reduction of physiological common mode signals for the VR though not for the FVR method (cf., panels (b)-(c) and (e)-(f) in Fig. 8). Moreover, the application of the FVR method showed a higher rejection of power line interference on all the channels of the detection system, regardless of how differently the power line interference affected the different channels in the detection system and on whether muscle activity was detected along the whole detection system or locally (cf., panels (a)-(c) and (d)-(f) in Fig. 8, respectively). The FVR method also preserved the physiological common mode signals, the standing waves, as indicated by the presence of nonpropagating potentials in Figs. 6 and 8(c) and (f). Overall, the FVR method proved useful to remove power line interference without affecting both the standing waves and the MUAPs present in the actual, monopolar EMGs.

V. Conclusion

In this paper, we propose a new method for the reduction of power line interference from monopolar EMG signals recorded with high-density detection systems. This method is an improvement of the VR technique, whose potential negative effect on the information content of monopolar signals has been demonstrated theoretically and experimentally. According to our key results, the FVR method seems to attenuate the power line interference minimizing the distortions of the actual, monopolar EMGs.

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